Cradle: An Open, Source-Cited Calculator for the Embodied Carbon of Indian Building Materials

Thesis submitted to the
Indian Institute of Technology, Kharagpur
In partial fulfillment of the requirements for the award of the degree of

B.Tech in Civil Engineering

By
Durgesh Kumar
Roll No.: 23CE36005

Under the guidance of
Dr. Vihangraj V. Kulkarni



CIVIL ENGINEERING DEPARTMENT
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
NOVEMBER 2026

CIVIL ENGINEERING DEPARTMENT
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

CERTIFICATE

This is to certify that we have examined the thesis entitled **Cradle: An Open, Source-Cited Calculator for the Embodied Carbon of Indian Building Materials**, submitted by **Durgesh Kumar** (Roll Number: 23CE36005) an undergraduate student of **Department of Civil Engineering** in partial fulfillment for the award of degree of B.Tech in Civil Engineering. We hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in partial fulfillment for the Under Graduate Degree for which it has been submitted. The thesis has fulfilled all the requirements as per the regulations of the Institute and has reached the standard needed for submission.

Date: November, 2026
Place: IIT Kharagpur

Dr. Vihangraj V. Kulkarni
Department of Civil Engineering
IIT Kharagpur

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Dr. Vihangraj V. Kulkarni, Department of Civil Engineering, IIT Kharagpur, for giving me the opportunity to work on this project and for the guidance, feedback and encouragement throughout it. This guidance shaped the direction of the work and helped me keep it focused on a problem that matters for Indian construction.

I am also grateful to Dr. Unnikrishna Menon, Postdoctoral Researcher, IIT Kharagpur, for the support and guidance on life cycle assessment. The discussions on LCA helped me understand the topic and build the calculator on a sound method.

I thank the Department of Civil Engineering, IIT Kharagpur, for the opportunity to carry out this project, and my friends and classmates for their suggestions and feedback. Finally, I thank my parents and family for their constant support and encouragement.

Durgesh Kumar
23CE36005

ABSTRACT

Embodied carbon is the greenhouse gas released while making, transporting and installing building materials. In India it is usually calculated by hand in spreadsheets, often with foreign or undocumented emission factors, so published results are hard to check or repeat.

This project builds **Cradle**, a free web calculator for the embodied carbon (global warming potential, GWP) of Indian building materials. It follows ISO 14040/14044 and the life-cycle modules of EN 15804: product (A1–A3), transport to site (A4), construction (A5), end of life (C1–C4) and benefits beyond the life cycle (D). It uses only free, publicly available data: 106 factors for Indian products, 54 of them from Indian Environmental Product Declarations (EPDs), each cited to the table or page it was read from. Material quantities for common work items come from the CPWD Analysis of Rates 2014.

The calculator was checked in two ways. Forty-nine automated tests check the arithmetic, the unit conversions and the data. It also reproduced a published Indian building study (Chaudhary and Akhtar, 2023): the product stage (A1–A3) matched exactly at 899,412.37 kg CO_2e, and construction (A5) to within 0.41 kg. A first application compares 1 m³ of masonry: clay brick gives 665 kg CO_2e, fly ash–lime brick 361 and AAC block 342. Changing only the brick-kiln factor moves the clay-brick result from 542 to 973 kg CO_2e/m³, which shows why every factor must be reported with its source. Future work includes more validation cases, a comparison of walls that do the same job, and wider uncertainty ranges.

Keywords: embodied carbon, life cycle assessment, global warming potential, EN 15804, Indian construction materials, EPD, CPWD, open data

CONTENTS

1 Introduction **1**

1.1 Background 1

1.2 The problem 1

1.3 Aim and objectives 2

1.4 Scope 2

1.5 Organisation of the thesis 2

2 Literature Review **3**

2.1 Life cycle assessment in brief 3

2.2 Indian studies 3

2.3 Where Indian factors come from 4

2.4 How much results vary 5

2.5 Can published results be checked? 5

2.6 Existing tools 6

2.7 Research gap 6

3 Methodology **7**

3.1 Overall workflow 7

3.2 Goal and scope 7

3.3 Emission factor library 7

3.4 Calculation 9

3.5 Material quantities from CPWD 9

3.6 Uncertainty 10

3.7 How much of a result is Indian data? 11

3.8 Entering the inventory 11

3.9 Software and testing 12

4 Results and Discussion **13**

4.1 The calculator 13

4.2 Validation against a published study 17

4.3 First application: 1 m^3 of masonry 18

4.4 How much does the brick factor matter? 18

4.5 Cement type 19

5 Conclusion and Future Works **20**

5.1 Conclusions 20

5.2 Limitations 20

5.3 Future Works 21

References **22**

LIST OF FIGURES

Figure 2.1	Life-cycle modules of EN 15804 and the boundary used in this project	3
Figure 2.2	Spread of the same input between Indian sources (highest value ÷ lowest value)	5
Figure 3.1	Workflow of the calculator, following the four steps of ISO 14044	7
Figure 3.2	The 106 factors in the library, by material category	8
Figure 3.3	AI draft of a brick wall: the model sizes the wall; CPWD coefficients give the materials, with the working on each line	12
Figure 4.1	Home page of Cradle with four saved projects	13
Figure 4.2	Ledger of the validated building (Section 4.2): 78 material lines with quantity, data tag, result and running total	14
Figure 4.3	Results of the validated building: 911 t CO_2e in total, the biggest lines and the Indian-data share	15
Figure 4.4	Compare screen: 1 m³ of clay brick, AAC block and fly ash–lime brick masonry on one scale	15
Figure 4.5	Report screen: an ISO 14044-style report of the validated building	16
Figure 4.6	Library screen: Indian cement factors, each with its EPD, page and data tag	16
Figure 4.7	A1–A3 emissions of the validated building by material (total 899.4 t CO_2e)	18
Figure 4.8	Embodied carbon of 1 m³ of three masonry types, split by ingredient (kg CO_2e, A1–A3)	18
Figure 4.9	1 m³ of clay brickwork for each IFC kiln factor (kg CO_2e; brick factor in kg CO_2e/kg in brackets)	19
Figure 4.10	Embodied carbon of Indian cement types from manufacturer EPDs (dot: mean; line: lowest to highest EPD)	19

LIST OF TABLES

Table 1.1	Objectives of the project and where each is addressed	2
Table 1.2	Scope of the project	2
Table 2.1	Indian embodied-carbon studies read in full	4
Table 2.2	Reproducibility check of Indian studies	5
Table 2.3	Tools used in Indian studies	6
Table 3.1	Data-quality tags and number of material factors	8
Table 3.2	Main transport and site-energy factors	8
Table 3.3	CPWD work items used in the calculator	10
Table 3.4	Conversions from CPWD quantities to kg	10
Table 3.5	Ways to enter the inventory	11
Table 3.6	Automated tests (all 49 pass)	12
Table 4.1	Main screens of Cradle	13
Table 4.2	Validation result	17
Table 5.1	Limitations of the present work	20

ABBREVIATIONS

AAC	Autoclaved aerated concrete
A1–A3, A4, A5, C1–C4, D	EN 15804 life-cycle modules (Figure 2.1)
BoQ	Bill of quantities
CEA	Central Electricity Authority
CO_2e	Carbon dioxide equivalent
CPWD	Central Public Works Department
DAR	Delhi Analysis of Rates (CPWD)
EF	Emission factor
EPD	Environmental Product Declaration
FaL-G	Fly ash–lime–gypsum (brick)
GWP	Global warming potential, in kg CO_2e
ICMD	IFC India Construction Materials Database
IFC	International Finance Corporation
LCA	Life cycle assessment
OPC / PPC / PSC	Ordinary Portland / Portland pozzolana / Portland slag cement

1. INTRODUCTION

1.1 Background

Buildings produce about 30% of India's greenhouse gas emissions, and 48% of building emissions are *embodied*: they come from making building materials, not from running the building [NITI Aayog, 2026]. In the building studied in Chapter 4, bricks, cement and steel alone cause 84% of the material emissions (Figure 4.7).

India is starting to ask for these numbers:

- **ECSBC 2024, Appendix A6** asks buildings to report embodied carbon in kg CO_2e per m² for stages A1–A3. It is voluntary for now [BEE, 2024].
- **GRIHA v2019, Criterion 20** gives up to 5 points for lowering a building's global warming potential, using an LCA tool and EPDs [GRIHA Council, 2019].
- **IGBC Net Zero Carbon** sets a design-stage limit of 700 kg CO_2e/m² [IGBC, n.d.].
- **Carbon Credit Trading Scheme (2023)** sets emission-intensity targets for 186 cement plants [Ministry of Power, 2023; GHG Intensity Target Rules, 2025].

So designers, students and researchers increasingly need an embodied-carbon figure they can defend.

1.2 The problem

Indian studies calculate embodied carbon in two ways. Some use paid LCA software (SimaPro with the European ecoinvent database). Many use plain spreadsheets [Chaudhary and Akhtar, 2023; Akshatha et al., 2025]. The literature review (Chapter 2) found four recurring problems:

1. **Foreign or undocumented data.** Factors often come from European databases or unnamed "literature" [Bhagat and Savoikar, 2021; Patil et al., 2023; Talpur et al., 2023].
2. **Large spread.** The factor for fired clay brick varies about 6 times between sources, from 0.097 to 0.59 kg CO_2e/kg (Figure 2.2).
3. **Results that cannot be checked.** Of 9 Indian studies that printed enough data to recompute, only 2 reproduced exactly (Table 2.2).
4. **No uncertainty.** None of the studies in Table 2.2 gives a range on its result. Among the Indian studies reviewed, only Tibrewal et al. (2023) did.

1.3 Aim and objectives

The aim is a free calculator that gives the embodied carbon of Indian building materials, in which every number can be traced to a source and checked by hand. Table 1.1 lists the objectives.

No.	Objective	Addressed in
1	Build an open library of emission factors for Indian materials, each cited to its table or page	Section 3.3 (106 factors)
2	Write a calculation engine for EN 15804 modules A1–A3, A4, A5, C1–C4 and D, with uncertainty	Sections 3.4 and 3.6
3	Validate the engine by reproducing a published Indian study	Section 4.2
4	Apply it to common Indian masonry and cement types, and test how results change with the choice of factor	Sections 4.3 to 4.5

Table 1.1: Objectives of the project and where each is addressed

1.4 Scope

Included	Not included, and why
Building materials (cement, concrete, steel, bricks, blocks, glass, finishes)	Operational energy of the building: a separate field with its own tools
One impact: global warming potential (kg CO_2e)	Other impacts (acidification, water use): no open Indian data exists for them
EN 15804 modules A1–A3, A4, A5, C1–C4 and D	Use stage (B1–B7): needs a building's maintenance plan
Free, publicly available data only	Paid databases (ecoinvent, GaBi): others cannot check them

Table 1.2: Scope of the project

1.5 Organisation of the thesis

Chapter 2 reviews LCA methods and Indian studies. Chapter 3 explains how the calculator works. Chapter 4 presents the validation and the results. Chapter 5 gives the conclusions, the limitations and future work.

2. LITERATURE REVIEW

2.1 Life cycle assessment in brief

Life cycle assessment (LCA) adds up the environmental impact of a product over its life. ISO 14040 and ISO 14044 set the method in four steps. BIS has adopted both as Indian standards, IS/ISO 14040 and 14044 [BIS, 2009].

1. **Goal and scope:** what is studied, for which unit (for example, 1 m³ of wall), and which life stages count.
2. **Inventory:** how much of each material, fuel and transport is used.
3. **Impact assessment:** inventory × emission factors = impact.
4. **Interpretation:** which items matter most, and how certain the answer is.

For construction products, EN 15804 splits the life into modules (Figure 2.1). This project reports one impact, the global warming potential (GWP), in kg of CO_2-equivalent (kg CO_2e). It adds all greenhouse gases using their 100-year warming effect relative to CO_2.

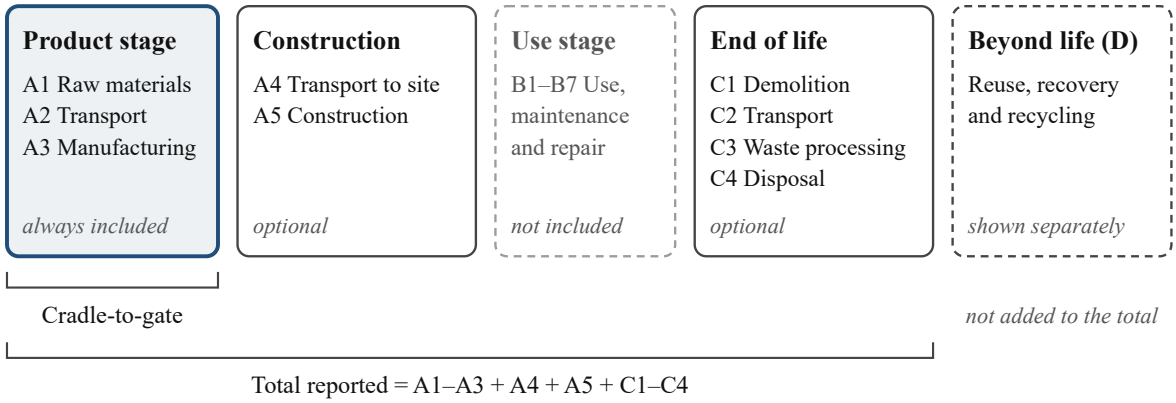


Figure 2.1: Life-cycle modules of EN 15804 and the boundary used in this project

2.2 Indian studies

Table 2.1 lists Indian studies that were read in full. Most stop at the factory gate (A1–A3). None of the material-level studies counted end of life (C1–C4).

Study	Topic	How calculated	Main result
Gettu et al. 2016	Indian cements	SimaPro + ecoinvent + plant data	OPC 870, PPC 660, PSC 730 kg CO_2/t
Gettu et al. 2018	30 concrete mixes	SimaPro + plant data	OPC 930, PPC 690, PSC 550 kg CO_2e/t
Pillai et al. 2019	OPC, fly ash and LC3 concrete	Plant data + ecoinvent	M35 OPC concrete 335 kg CO_2e/m³
Patil et al. 2023	Geopolymer vs OPC concrete	Spreadsheet	306.8 vs 373.9 kg CO_2e/m³
Singh et al. 2024	Clay vs fly ash brick	Hand calculation	290 vs 43 g CO_2 per brick
Tibrewal et al. 2023	Brick kiln energy, 500+ kilns	Field survey	Zigzag kilns use the least energy
IFC 2017 (ICMD)	National factor database	GaBi software	OPC 0.91; clay brick 0.32–0.59 kg CO_2e/kg
Chaudhary & Akhtar 2023	Residential building, Bihar	Excel	936 t CO_2e (A1–A5)
Akshatha et al. 2025	30-storey tower	Spreadsheet, 4 data sets	373 to 530 kg CO_2e/m²
AEEE 2024	Office building	One Click LCA vs IFC data	Results differ by about 17%

Table 2.1: Indian embodied-carbon studies read in full

2.3 Where Indian factors come from

Three sources dominate, and each has a weakness.

- **ecoinvent** (European, paid): a review of 71 South Asian building LCAs found it was always the database used [Talpur et al., 2023]. It is built on European production.
- **IFC India Construction Materials Database (ICMD), 2017** [IFC, 2017]: the main Indian source. It is dated (2017, with 2012 grid electricity), gives only average values, and is marked "draft".
- **Indian EPDs:** verified product data from manufacturers. They are still few; India had about 1.7% of the EPDs the International EPD System published in 2025 [International EPD System, 2025].

The sources can also disagree. ICMD ranks zigzag-kiln bricks as the worst kiln type (0.59 kg CO_2e/kg). A survey of more than 500 kilns found that zigzag kilns use the *least* firing energy, 1.19 MJ/kg against 1.30 MJ/kg for bull's trench kilns [Tibrewal et al., 2023; IFC, 2017].

2.4 How much results vary

The same input can differ several times between Indian sources (Figure 2.2). The choice of factor can therefore change a result more than the design choice being studied.

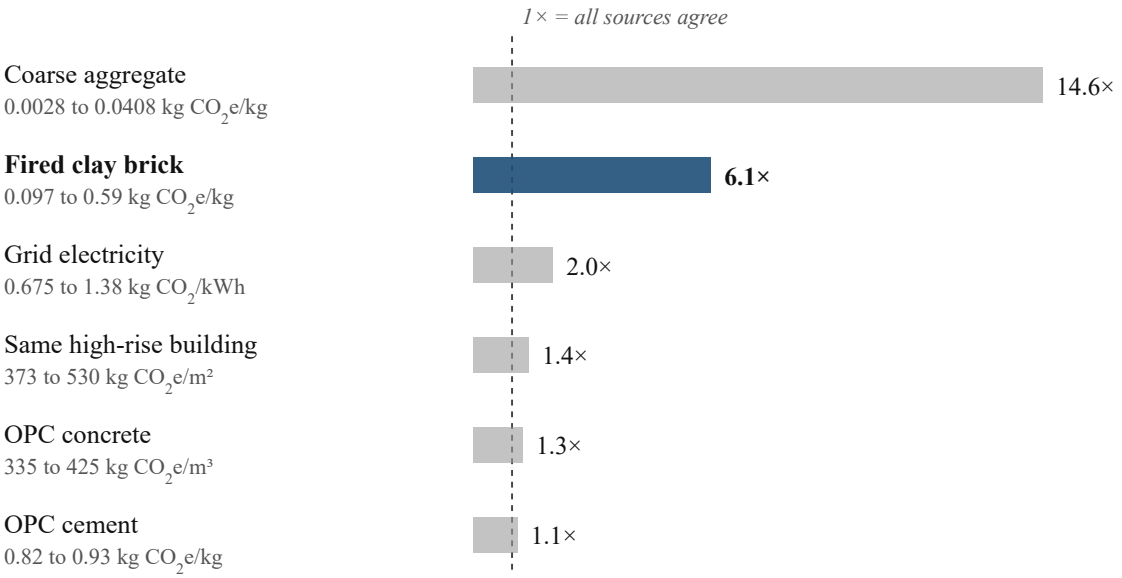


Figure 2.2: Spread of the same input between Indian sources (highest value ÷ lowest value)

2.5 Can published results be checked?

The arithmetic of every Indian study that printed its inventory and factors was redone (Table 2.2).

Study	Inventory printed?	Does the result reproduce?	Uncertainty?
Chaudhary & Akhtar 2023	Yes, in full	Yes: 936.1 t vs 936 t	No
Patil et al. 2023	Yes	Yes, exactly	No
Bhagat & Savoikar 2021	Yes	About 97% (transport not given)	No
Pillai et al. 2019	Yes	Ranking yes; values 42–48 kg/m³ higher	No
Singh et al. 2024	Partly	No: 97.08 g/kg × 2.8 kg = 271.8 g, but 290 g printed	No
Ansari et al. 2026	Yes	No: formula gives 386.88, paper states 302.64	No
Padavala et al. 2026	Yes	No: cement alone (382.5) exceeds the total (368)	No
Omran et al. 2026	Yes	Cannot test: no factors printed	No
Kumbhar et al. 2014	Yes	No kg CO_2e reported	No

Table 2.2: Reproducibility check of Indian studies

Only Chaudhary and Akhtar printed a full bill of quantities, a source for every factor, and results by stage. This makes it the best case for validating a calculator (Section 4.2).

2.6 Existing tools

Tool	What it is	Limitation for Indian researchers
SimaPro [Gettu et al., 2016; Pillai et al., 2019]	Commercial LCA software, usually with ecoinvent	Paid; background data mostly European
openLCA [Kar et al., 2026]	Free LCA software	Needs a database; the 395 Indian datasets that exist sit inside paid databases [Siluvai Antony and Jayaram, 2019]
One Click LCA [AEEE, 2024]	Commercial building LCA tool	Foreign factors adjusted to the Indian grid
EDGE (IFC) [IFC, 2017]	Green-building app that hosts the ICMD	ICMD gives averages only, last updated 2017
Spreadsheets [Chaudhary and Akhtar, 2023]	Hand-built Excel sheets	Every study rebuilds the sheet; errors are hard to spot (Table 2.2)

Table 2.3: Tools used in Indian studies

2.7 Research gap

No free tool combines (a) Indian factors that are each cited to a page, (b) working shown for every number, (c) a check against a published study, and (d) an uncertainty range. Several Indian reports ask for such open data: the CSIR-NEERI and CII national LCA database roadmap [Siluvai Antony and Jayaram, 2019], CEPT University's database framework [Rawal et al., 2024] and NITI Aayog's call for a public EPD register [NITI Aayog, 2026]. This project addresses the gap for one impact (GWP) and one sector (building materials).

3. METHODOLOGY

3.1 Overall workflow

Cradle follows the four ISO 14044 steps (Figure 3.1). The user sets the goal and boundary and enters the materials; the engine multiplies each quantity by a cited factor. Results come with their working, an uncertainty range and a report.

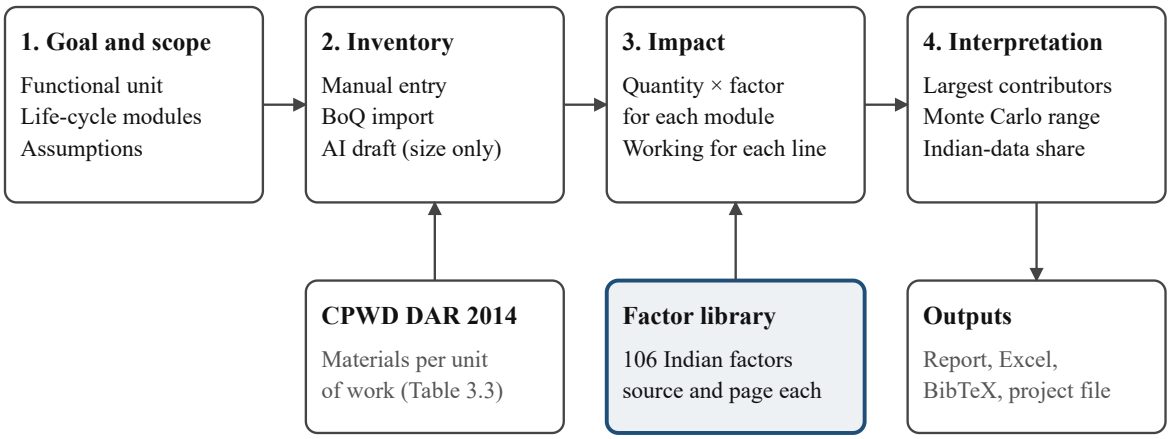


Figure 3.1: Workflow of the calculator, following the four steps of ISO 14044

3.2 Goal and scope

Every project starts with three choices, which are saved with the project and printed in its report:

- **Functional unit:** what the result refers to, such as "1 m³ of masonry" or "whole building". If a floor area is given, the result is also shown per m².
- **System boundary:** which EN 15804 modules to include (Figure 2.1). A1–A3 is always on; A4, A5, C1–C4 and D can be switched on.
- **Assumptions:** free text, for example the source of the quantities.

3.3 Emission factor library

The library holds 106 A1–A3 factors, all for Indian products (Figure 3.2). Four rules were followed:

1. **Free data only.** No ecoinvent, GaBi or other paid data, so anyone can check a value.
2. **Indian first.** Indian EPDs first, then Indian studies and Government of India data.
3. **Read from the document.** Each value was read from its PDF or spreadsheet, never from memory.
4. **Cited to a page.** Each factor stores its source, the table or page, the year and the licence.

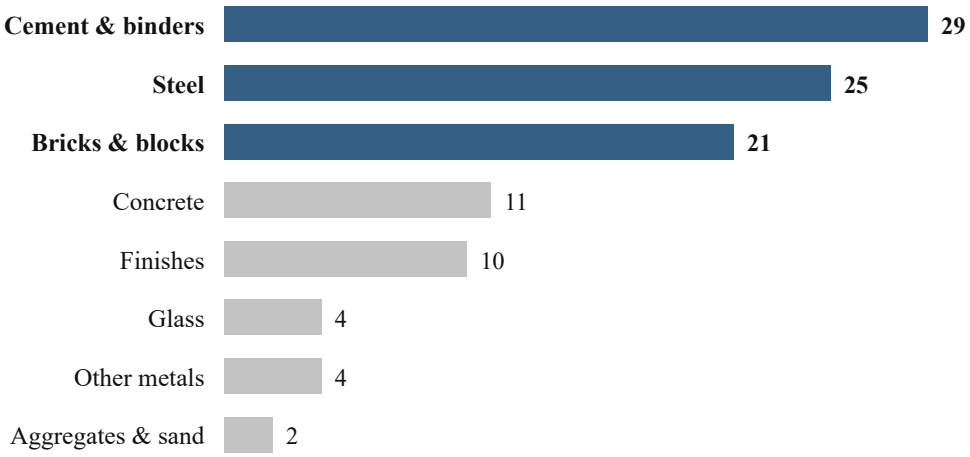


Figure 3.2: The 106 factors in the library, by material category

Each factor carries a data-quality tag (Table 3.1). The results use it to show how much of a total rests on Indian data (Section 3.7).

Tag	Meaning	Factors
Indian EPD	Third-party verified declaration of a product made in India	54
Indian study	Peer-reviewed or technical study of Indian production (e.g. IFC 2017)	45
Govt. of India	Published by a government body (e.g. Ministry of Steel 2024)	6
Live India data	Plant-level data fetched from Climate TRACE	1
Total		**106**

Table 3.1: Data-quality tags and number of material factors

Transport (A4) and site energy (A5) use separate factors (Table 3.2). Indian values are used where they exist; UK values (DESNZ 2025, open licence) fill the gaps and are labelled as proxies.

Factor	Value	Unit	Source
Heavy truck, >12 t, diesel (India)	0.7375	kg CO_2e per vehicle-km	India GHG Program 2015, p.10
Medium truck, <12 t, diesel (India)	0.5928	kg CO_2e per vehicle-km	India GHG Program 2015, p.10
Rail freight, Indian Railways	0.00996	kg CO_2e per t-km	India GHG Program 2015, p.7
Grid electricity, India FY2025-26	0.6751	kg CO_2 per kWh	CEA CO_2 Baseline Database v22
Diesel burned on site	2.66155	kg CO_2e per litre	DESNZ 2025 (UK proxy)

Table 3.2: Main transport and site-energy factors

3.4 Calculation

The engine is linear: every number is a quantity times a factor. For each material line i:

$$\text{A1–A3}_i = Q_i \times EF_i \tag{3.1}$$

where Q is the quantity in the factor's declared unit (kg, t, m³ or piece) and EF is the factor. If the user enters a different unit, the engine converts it with the factor's density or unit mass. If no conversion exists, it shows a warning instead of guessing.

Transport to site (A4) uses the mass m (t) and the distance d (km). Rail and ship factors are per tonne-km (Equation 3.2). Truck factors are per vehicle-km, so the mass is first turned into trips using the truck's payload P in tonnes (Equation 3.3):

$$\text{A4}_i = m_i \times d \times EF_{\text{t·km}} \tag{3.2}$$

$$\text{A4}_i = (m_i \div P) \times d \times EF_{\text{vehicle·km}} \tag{3.3}$$

Construction (A5) counts the wasted share w of each material, plus the energy E used on site (electricity, diesel, water):

$$\text{A5} = \Sigma\, w_i \times (\text{A1–A3}_i + \text{A4}_i) + \Sigma\, E_j \times EF_j \tag{3.4}$$

End of life (C1–C4) uses the product's own EPD value when it declares one; otherwise the mass in tonnes times a route factor for landfill, recycling or incineration:

$$\text{C1–C4}_i = m_i \times EF_{\text{route}} \tag{3.5}$$

Module D (benefits from recycling) is shown separately and never added to the total, as EN 15804 requires:

$$\text{Total} = \text{A1–A3} + \text{A4} + \text{A5} + \text{C1–C4} \tag{3.6}$$

For every line the engine also writes out its working, for example "1,540 kg × 0.39 kg CO_2e/kg = 600.5 kg CO_2e". Any number can therefore be checked by hand.

3.5 Material quantities from CPWD

For common work items, the materials per unit of work come from the CPWD Delhi Analysis of Rates (DAR) 2014 (Table 3.3). Volumes and brick counts are converted to kg with densities from IS 875 (Part 1), IS 1077 and IFC 2017 (Table 3.4).

Work item (per m³)	Materials per m³ of work	CPWD source
Clay brickwork, mortar 1:6	494 bricks + 0.25 m³ mortar	DAR 2014 item 6.4.2, p.234
Fly ash–lime (FaL-G) brickwork, mortar 1:6	487 bricks + 0.22 m³ mortar	DAR 2014 item 6.34.2, p.255
AAC block masonry, mortar 1:4	1 m³ blocks + 0.15 m³ mortar + 13.2 kg steel	DAR 2014 item 6.38, pp.257–258
M25 design-mix concrete	0.33 t cement + 0.425 m³ sand + 0.85 m³ aggregate	DAR 2014 item 5.33.1, p.207

Table 3.3: CPWD work items used in the calculator

Conversion	Value	Source
Mortar 1:6, per m³	0.25 t cement + 1.07 m³ sand	DAR 2014 item 3.11, p.119
Mortar 1:4, per m³	0.38 t cement + 1.07 m³ sand	DAR 2014 item 3.9, p.118
Sand density	1,570 kg/m³ (midpoint of 1,540–1,600)	IS 875 (Part 1), Table 1
Coarse aggregate density	1,735 kg/m³ (midpoint of 1,600–1,870)	IS 875 (Part 1), Table 1
Brick mass	230 × 110 × 70 mm × 1,760 kg/m³ = 3.12 kg	IS 1077; IFC 2017, Table 18
AAC density	500 kg/m³	IFC 2017, Table 18

Table 3.4: Conversions from CPWD quantities to kg

Two tests protect these numbers. The cement in each item matches CPWD's separately published cement coefficients (DSR 2014). The mass of 1 m³ of clay brickwork comes out within 10% of the 1,920 kg/m³ given in IS 875.

3.6 Uncertainty

Where a source gives a range (low to high) for a factor, the engine runs a Monte Carlo simulation:

1. Each ranged factor is drawn from a triangular distribution: lowest at the low value, peak at the stated value, back to zero at the high value.
2. The project is recalculated with the drawn values. This is repeated 5,000 times.
3. The 2.5th and 97.5th percentiles of the 5,000 totals give a 95% range.

The random generator uses a fixed seed (42), so the same project always gives the same range. Factors without a range stay fixed, and the result states what share of the total was varied. At present only 9 of the 106 factors have a published range (for example, the mean

of four Indian OPC EPDs, 921 to 1,010 kg CO_2e/t). The ranges are therefore narrow; Section 5.2 treats this as a limitation.

3.7 How much of a result is Indian data?

The engine splits each total by the quality tag of the factor behind every kilogram. A result can then say, for example, "83% rests on Indian studies, 17% on foreign proxies". This makes the strength of the data visible next to the number.

3.8 Entering the inventory

There are three ways to enter materials (Table 3.5). All three end in the same review table, where the user confirms each line.

Method	How it works	Who decides the numbers
Manual entry	Pick a material from the library and type the quantity and unit	The user
BoQ import	Upload a CPWD-style bill of quantities (CSV or Excel). Units such as cum, sqm, MT, qtl, bags and nos are read, and each row is matched to a library material	The BoQ; the user confirms each match
AI draft	Type or speak a short description, e.g. "230 mm brick wall, 10 m²". A language model (Groq, gpt-oss-120b) works out only the size of the work item (10 m² × 0.23 m = 2.3 m³). The CPWD coefficients in Table 3.3 turn it into materials	The model gives only the size; materials and factors come from the cited tables

Table 3.5: Ways to enter the inventory

The AI is never allowed to supply an emission factor or a material quantity per unit of work. In an early test where it did, the brick and mortar quantities were wrong by about 2 times.

Figure 3.3 shows a real draft for "230 mm thick clay brick wall in cement mortar 1:6, 3 m high and 4 m long". The model gave only the size (3 m × 4 m × 0.23 m = 2.76 m³). The bricks, cement and sand then follow from CPWD DAR 2014 item 6.4.2, and each line shows this working before the user adds it to the ledger.

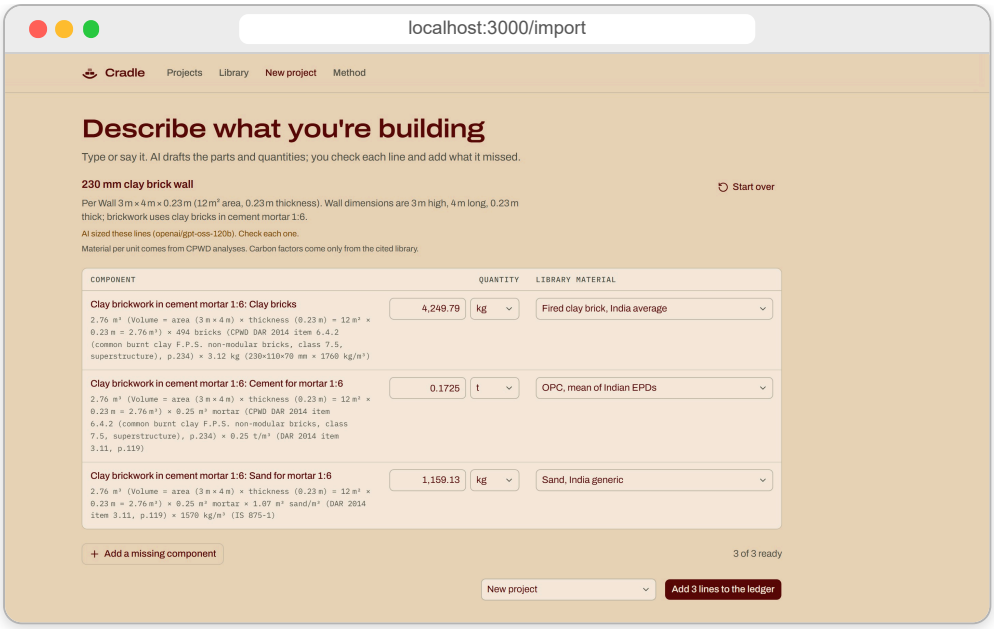


Figure 3.3: AI draft of a brick wall: the model sizes the wall; CPWD coefficients give the materials, with the working on each line

3.9 Software and testing

Cradle is a web app built with Next.js and TypeScript. The calculation engine is separate from the user interface, so it can be tested on its own. Projects are saved in the user's browser and can be exported as a project file, an Excel workbook with live formulas, a printable report and a BibTeX reference list. Table 3.6 lists the automated tests.

Test file	What it checks	Tests
Engine	Each module's equation, unit conversion, boundary switches, D kept out of the total, Monte Carlo	14
AI draft	The model's answer is checked and expanded only into cited work items	9
CPWD work items	Quantities match CPWD's cement table; brickwork mass within 10% of IS 875	7
BoQ import	Column detection and Indian BoQ units	6
Library	Every factor has a source, a page and a licence; ranges contain the value	5
Validation	Published study reproduced (Section 4.2)	3
Excel export	Every formula in the workbook gives the engine's value	3
Number input	Typed numbers are read correctly	2
Total		**49**

Table 3.6: Automated tests (all 49 pass)

4. RESULTS AND DISCUSSION

4.1 The calculator

Cradle works, and all 49 tests pass (Table 3.6). Table 4.1 lists its main screens. The figures below are screenshots of the running app with real data: the three masonry walls of Section 4.3 and the validated building of Section 4.2.

Screen	What the user does there	Figure
Home	Opens a worked example or a saved project; describes a new project for the AI draft	4.1
Ledger	Adds materials, transport, wastage and site energy; sees a running total line by line	4.2
Results	Sees totals by module, the largest contributors, the uncertainty range and the Indian-data share	4.3
Compare	Puts alternatives (e.g. three wall types) on one scale	4.4
Report	Prints an ISO 14044-style report; downloads Excel, BibTeX or the project file	4.5
Library	Browses the 106 factors, each with its source and page	4.6

Table 4.1: Main screens of Cradle

The home page (Figure 4.1) lists the projects saved in the browser with their totals. A one-line description starts an AI draft, and worked examples open with one click.

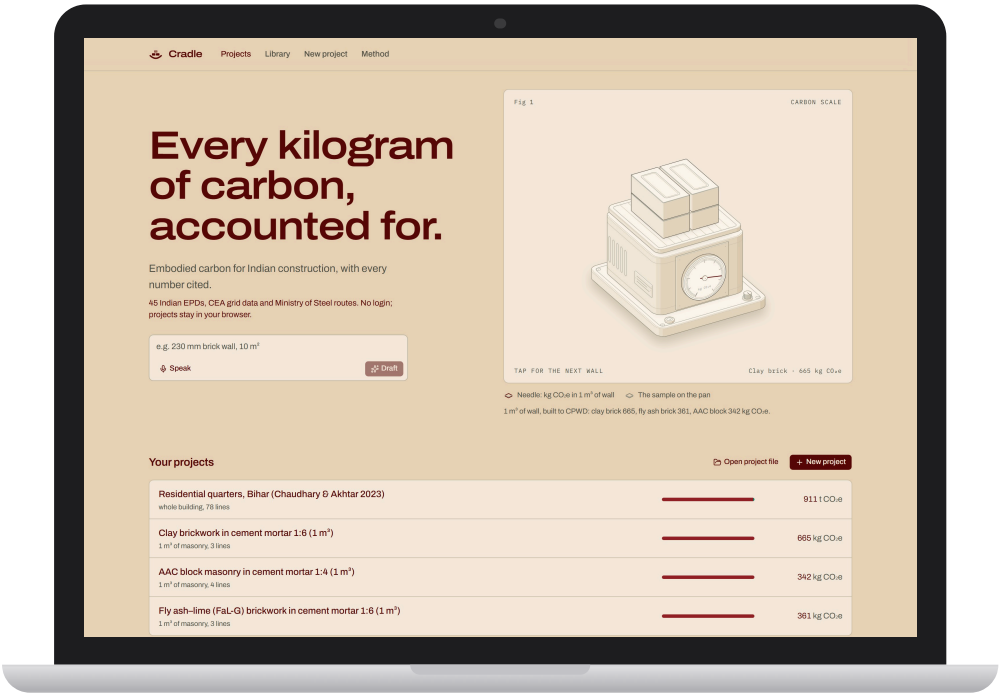


Figure 4.1: Home page of Cradle with four saved projects

The ledger (Figure 4.2) is the main working screen. Each line shows the material, its quantity and unit, a tag for the kind of data behind its factor ("Indian study", "Proxy" and so on), its result and a running total; opening a line shows its working and its source. The results screen (Figure 4.3) gives the total, the split by module, the biggest lines and the share of the total that rests on Indian data. For the validated building the total is 911 t CO_2e (A1–A3 and A5), and 83.3% of it rests on Indian data.

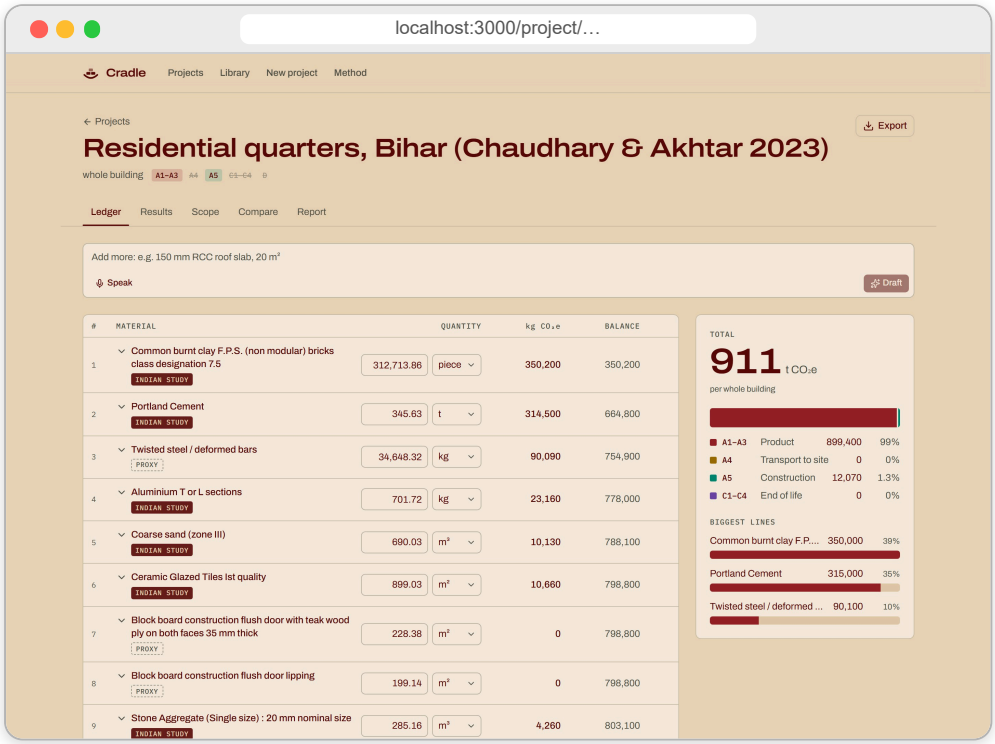


Figure 4.2: Ledger of the validated building (Section 4.2): 78 material lines with quantity, data tag, result and running total

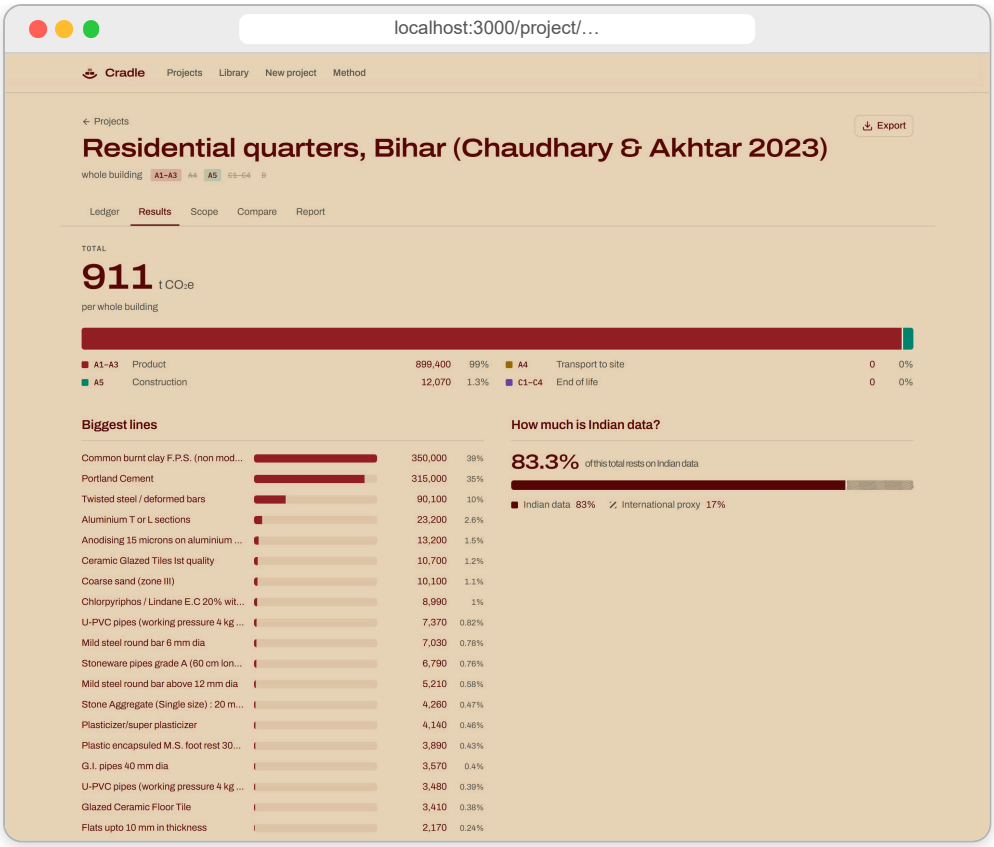


Figure 4.3: Results of the validated building: 911 t CO_2e in total, the biggest lines and the Indian-data share

The compare screen (Figure 4.4) puts saved projects on one scale. Here the three masonry walls of Section 4.3 are compared: AAC and fly ash–lime brick masonry come out 49% and 46% below clay brick masonry. The report screen (Figure 4.5) prints the project in the order ISO 14044 asks for: goal, scope, inventory with a source for every factor, results and interpretation. The library (Figure 4.6) lists every factor with its source, page and data tag, so a user can check any value before using it.

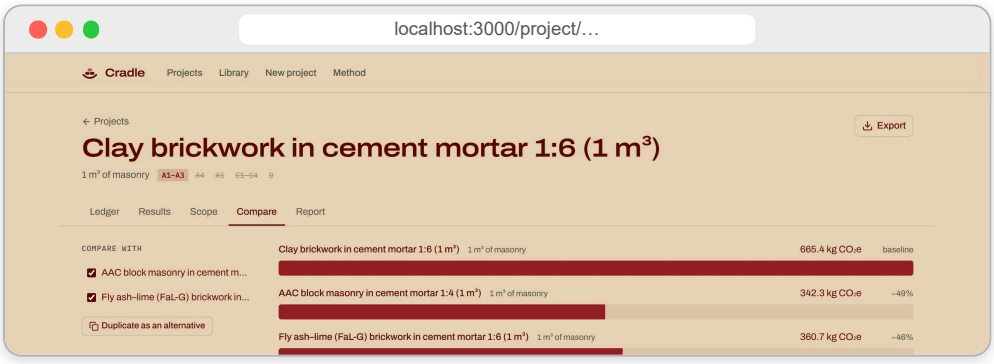


Figure 4.4: Compare screen: 1 m³ of clay brick, AAC block and fly ash–lime brick masonry on one scale

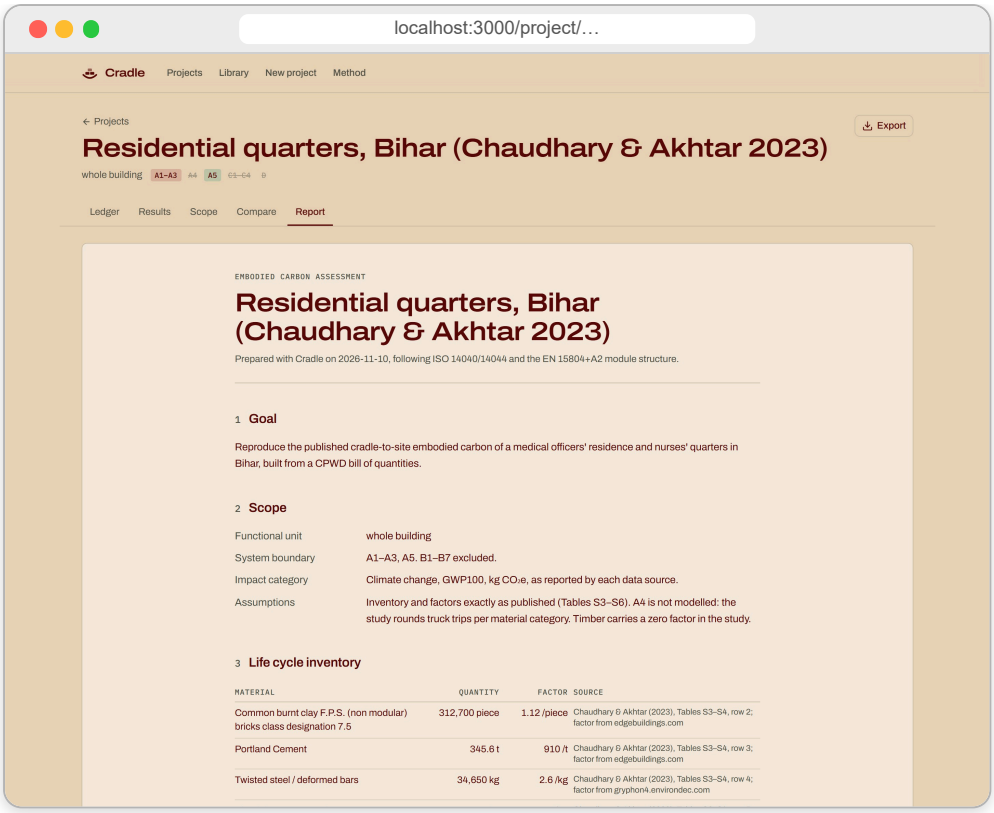


Figure 4.5: Report screen: an ISO 14044-style report of the validated building

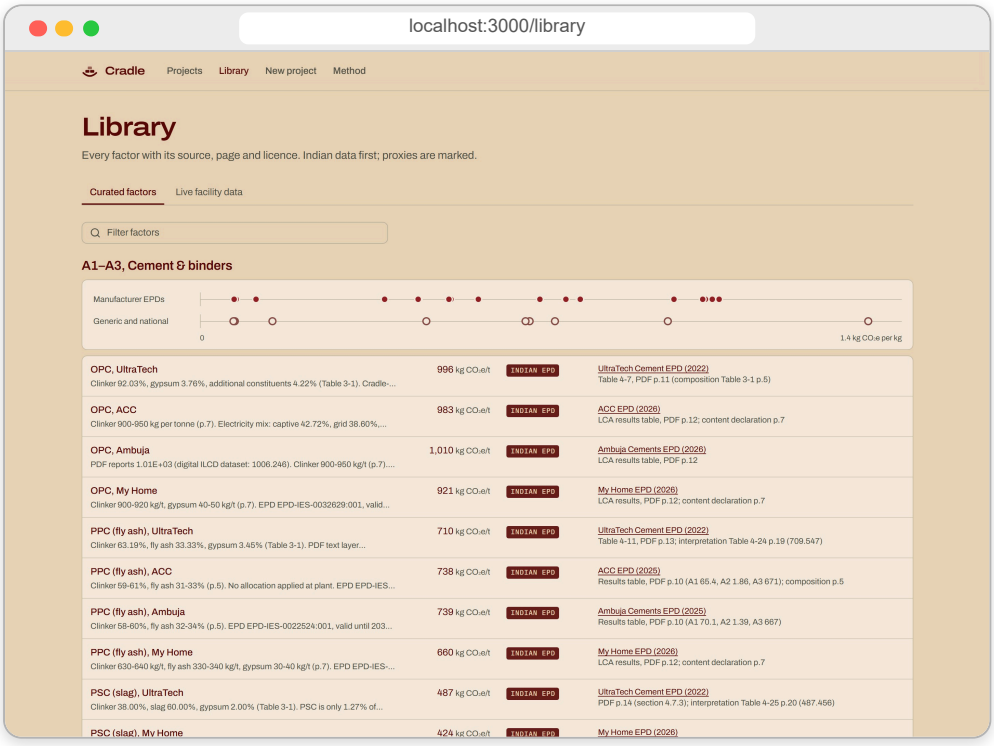


Figure 4.6: Library screen: Indian cement factors, each with its EPD, page and data tag

4.2 Validation against a published study

The engine was checked against Chaudhary and Akhtar (2023), who studied residential quarters at a hospital in Bhojpur, Bihar. They turned a CPWD bill of quantities into 78 materials and gave each an emission factor with its source.

Method. All 78 materials were entered with the paper's own quantities, units and factors. Because the data is the same, any difference can only come from the calculation itself: units, conversions or sums. The paper uses mixed units (1000 bricks, quintals, packs of 10 and 100, litres, metres), so the unit conversions were tested too. Table 4.2 gives the result.

Stage	Published (kg CO_2e)	Cradle (kg CO_2e)	Difference
A1–A3 material production	899,412.37	899,412.37	0.00
A5 on-site construction	12,075.01	12,074.60	0.41 kg (0.003%)
A4 transport	24,631	not compared	see below

Table 4.2: Validation result

- **A1–A3:** every one of the 78 lines matches the paper's Table S4 after unit conversion, and so does the total.
- **A5:** the 0.41 kg gap comes from one welding item that the paper gives per cm, a unit the calculator does not have.
- **A4:** the paper rounds truck trips up to whole trips for each material group, while Cradle shares trips in proportion to mass. A4 is 2.6% of the total and is left out of this check. Adding the paper's trip method is listed as future work (Section 5.3).

Figure 4.7 shows where the building's A1–A3 emissions come from. Bricks and cement together make up 74%, so their factors matter most. The data-share check (Section 3.7) finds that 83.3% of this total rests on Indian data (the IFC database) and 16.7% on foreign factors the paper used.

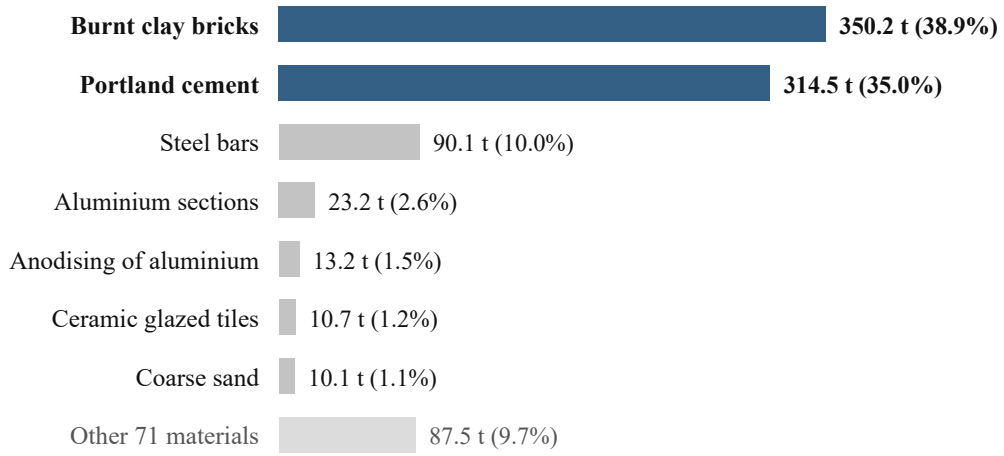


Figure 4.7: A1–A3 emissions of the validated building by material (total 899.4 t CO_2e)

4.3 First application: 1 m³ of masonry

Three common wall materials were compared using the CPWD quantities in Table 3.3 and factors from the library. The boundary is A1–A3 and the unit is 1 m³ of masonry (Figure 4.8).

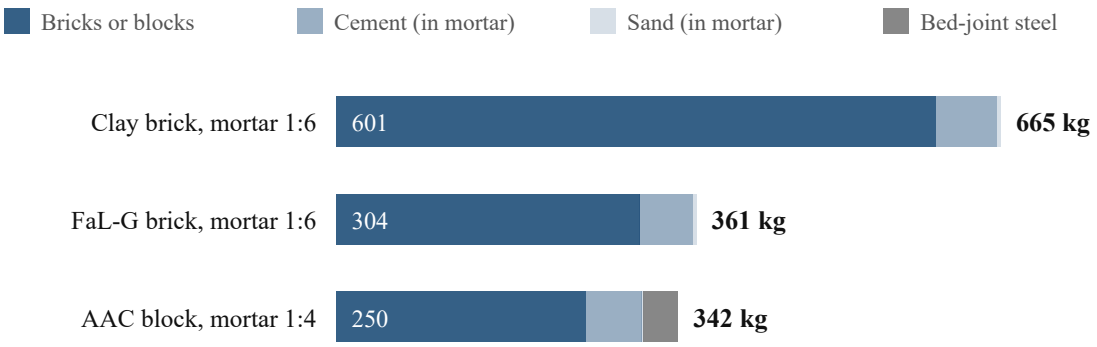


Figure 4.8: Embodied carbon of 1 m³ of three masonry types, split by ingredient (kg CO_2e, A1–A3)

Fly ash–lime brick and AAC block masonry come out at about half of clay brick masonry. The bricks or blocks themselves cause most of each total. Mortar cement adds 54–61 kg, and the AAC bed-joint steel adds 34 kg.

This comparison is per m³ of masonry. Walls of different materials differ in thickness, strength and thermal behaviour, so a design decision should compare walls that do the same job. That comparison is left for future work (Section 5.3).

4.4 How much does the brick factor matter?

The clay-brick result above uses the IFC India average (0.39 kg CO_2e/kg). IFC also gives a factor for each kiln type. Keeping the CPWD quantities fixed and changing only the brick factor gives Equation 4.1 and Figure 4.9:

$$\text{Wall} = 1{,}539.8 \text{ kg bricks} \times EF_{\text{brick}} + 64.9 \text{ kg } CO_2e \text{ (mortar)} \quad (4.1)$$

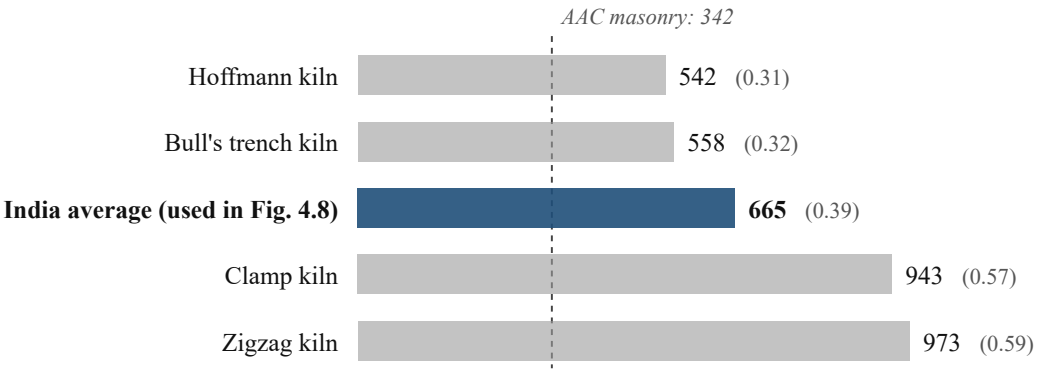


Figure 4.9: 1 m³ of clay brickwork for each IFC kiln factor (kg CO_2e; brick factor in kg CO_2e/kg in brackets)

The kiln assumption alone changes the result by 1.8 times (542 to 973 kg CO_2e/m^3). This is larger than the gap between clay brick and AAC in Figure 4.8. The data also conflict: IFC ranks the zigzag kiln worst, but a field survey of more than 500 kilns found that zigzag kilns use the least energy [Tibrewal et al., 2023]. A result for brick masonry should therefore always state which brick factor was used.

4.5 Cement type

Cement is the second-largest item in Figure 4.7. Indian EPDs give values for four cement types (Figure 4.10).

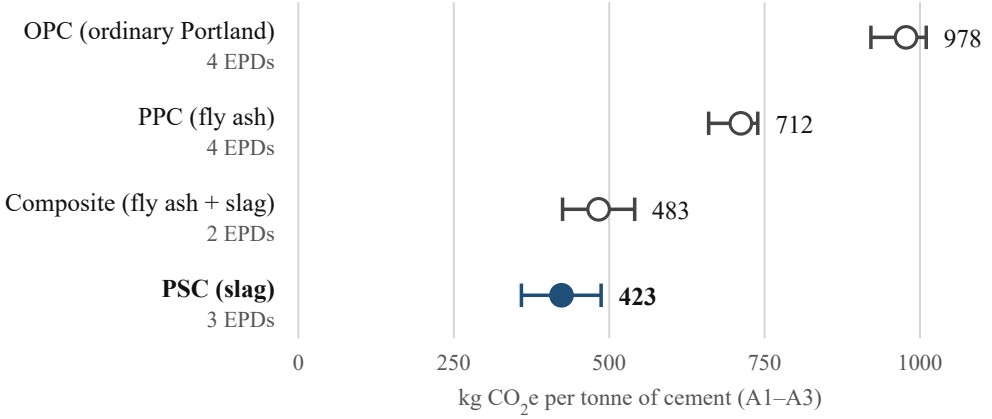


Figure 4.10: Embodied carbon of Indian cement types from manufacturer EPDs (dot: mean; line: lowest to highest EPD)

Slag cement (PSC) averages 423 kg CO_2e/t, less than half of OPC at 978. Blended cements replace part of the clinker, the high-carbon ingredient, with fly ash or slag. Even within one type the brands differ: OPC ranges from 921 to 1,010 kg CO_2e/t across four Indian EPDs. These EPD ranges are what the Monte Carlo analysis samples from (Section 3.6).

5. CONCLUSION AND FUTURE WORKS

5.1 Conclusions

1. A free calculator, Cradle, now computes the embodied carbon of Indian building materials for EN 15804 modules A1–A3, A4, A5, C1–C4 and D, and shows the working for every number.
2. Its library holds 106 factors for Indian products, each cited to its table or page; 54 of them come from Indian EPDs.
3. It reproduces a published Indian building study: A1–A3 exactly (899,412.37 kg CO_2e) and A5 within 0.41 kg.
4. For 1 m³ of masonry, fly ash–lime brick (361 kg CO_2e) and AAC block (342) come out at about half of clay brick (665).
5. The choice of brick factor alone moves the clay-brick result from 542 to 973 kg CO_2e/m³. Reporting the factor and its source is therefore as important as the result.

5.2 Limitations

Table 5.1 lists the limits of the present work and the step that would remove each one. Section 5.3 describes these steps.

Limitation	Effect	Next step
Only 9 of 106 factors have a published range	Uncertainty ranges are narrow; for walls they cover only 9–16% of the total	Add ranges wherever a source gives several values
Transport (A4) not yet validated	A4 results are not yet checked against a study	Add the paper's whole-trip method and re-check
Brick and block factors rely on IFC 2017	Dated averages; the kiln ranking conflicts with field data	Add Indian EPDs and field studies as they appear
One validation case	One building type only	Add a second validation case
Masonry compared per m³	Not walls that do the same job	Compare walls per m² of equal function
Licences	CEA asks for permission before public reuse; IFC and India GHG Program data are "all rights reserved"	Cite with attribution; ask CEA before public release
GWP only	Other impacts not covered	Out of scope: no open Indian data

Table 5.1: Limitations of the present work

5.3 Future Works

This work gives a working, validated calculator and a first set of results. The following steps would make it more reliable and more useful to Indian researchers.

1. **More validation cases.** The engine is checked against one building. Patil et al. (2023) printed every input for two concrete mixes, and its result reproduces exactly by hand, so it can be added as a second test. Pillai et al. (2019) uses Indian plant data; its ranking of mixes should match, and the constant gap of 42–48 kg/m³ (Table 2.2) should be explained.
2. **Transport to site (A4).** Add the whole-truck-trip method used by Chaudhary and Akhtar (2023), so that A4 can also be checked against their 24.6 t CO_2e.
3. **Wider uncertainty.** Only 9 of the 106 factors carry a range today. Where a source gives several values for one product, such as the four Indian EPDs for OPC, the range can be stored, so the Monte Carlo analysis covers more of each total.
4. **Walls that do the same job.** Section 4.3 compares 1 m³ of masonry. A fairer comparison is per m² of wall with the same function, for example a 230 mm clay brick wall against an AAC block wall of similar strength, with transport, wastage and end of life included.
5. **Sensitivity to method choices.** Test how results change with the brick kiln type (Section 4.4), the year of the CEA grid factor (0.7272 kg CO_2/kWh in FY2023-24 against 0.6751 in FY2025-26), and how much of a power plant's or steel plant's emissions is assigned to fly ash and slag, which are by-products.
6. **Indian end-of-life data.** End of life (C1–C4) now uses UK proxy factors (DESNZ 2025), because no open Indian dataset exists. Indian data on the recycling of construction and demolition waste, and on the reuse of fly ash and slag, would make modules C1–C4 and D more representative.
7. **More Indian EPDs.** As Indian manufacturers publish new EPDs, especially for bricks and blocks, they can replace the 2017 IFC averages in the library.

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